

DATA ENG 300 Homework 1: Aircraft Inventory Analysis

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import root_mean_squared_error
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import LabelEncoder
```

```
!curl -L -O https://github.com/mosesyhc/de300-2026sp/raw/refs/heads/main/homework/T_F41SCHEDULE_B43_with_missing
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current	
			Dload	Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	--:--:--	0
100	1577k	100	1577k	0	--:--:--	--:--:--	--:--:--	2985k

```
df = pd.read_csv('T_F41SCHEDULE_B43_with_missing.zip')
print(df.shape)
df.head()
```

(132313, 17)

/tmp/ipykernel_14600/3025076647.py:1: DtypeWarning: Columns (11) have mixed types. Specify dtype option on import
df = pd.read_csv('T_F41SCHEDULE_B43_with_missing.zip')

	YEAR	CARRIER	CARRIER_NAME	MANUFACTURE_YEAR	UNIQUE_CARRIER_NAME	SERIAL_NUMBER	TAIL_NUMBER	AIRCRAFT_STATUS
0	2006	16	PSA Airlines Inc.	2003.0	PSA Airlines Inc.	7858	N202PS	B
1	2006	16	PSA Airlines Inc.	2003.0	PSA Airlines Inc.	7860	N206PS	B
2	2006	16	PSA Airlines Inc.	2003.0	PSA Airlines Inc.	7873	N207PS	B
3	2006	16	PSA Airlines Inc.	2003.0	PSA Airlines Inc.	7874	N209PS	B
4	2006	16	PSA Airlines Inc.	2003.0	PSA Airlines Inc.	7879	N213PS	B

TASK 1: MISSING DATA INVESTIGATION AND IMPUTATION

```
cols_of_interest = ['CARRIER', 'CARRIER_NAME', 'AIRLINE_ID',
                    'MANUFACTURE_YEAR', 'NUMBER_OF_SEATS', 'CAPACITY_IN_POUNDS']

missing = df[cols_of_interest].isnull().sum()
pct = (missing / len(df) * 100).round(2)
pd.DataFrame({'Missing Count': missing, 'Missing %': pct})
```

	Missing Count	Missing %
CARRIER	49	0.04
CARRIER_NAME	105	0.08
AIRLINE_ID	105	0.08
MANUFACTURE_YEAR	3	0.00
NUMBER_OF_SEATS	13238	10.01
CAPACITY_IN_POUNDS	13332	10.08

1.1 CARRIER & CARRIER_NAME

```
# check all combinations for CARRIER / CARRIER_NAME
print("CARRIER null, CARRIER_NAME present:",
      df[df['CARRIER'].isnull() & df['CARRIER_NAME'].notna()].shape[0])
print("CARRIER present, CARRIER_NAME null:",
      df[df['CARRIER'].notna() & df['CARRIER_NAME'].isnull()].shape[0])
print("Both null:",
      df[df['CARRIER'].isnull() & df['CARRIER_NAME'].isnull()].shape[0])
```

```
CARRIER null, CARRIER_NAME present: 49
CARRIER present, CARRIER_NAME null: 105
Both null: 0
```

```
df[df['CARRIER'].isnull() & df['CARRIER_NAME'].notna()][['CARRIER', 'CARRIER_NAME', 'AIRLINE_ID']].drop_duplicates()
```

	CARRIER	CARRIER_NAME	AIRLINE_ID
11776	NaN	North American Airlines	20415.0

```
df[df['CARRIER'].notna() & df['CARRIER_NAME'].isnull()][['CARRIER', 'CARRIER_NAME', 'AIRLINE_ID']].drop_duplicates()
```

	CARRIER	CARRIER_NAME	AIRLINE_ID
11465	L4	NaN	NaN
54610	OH	NaN	NaN

Sometimes either CARRIER or CARRIER_NAME is missing while the other is available. Since each carrier name maps to a specific code, I use the existing data to fill in the missing values. This isn't statistical imputation, just a straightforward lookup, and I apply the same idea in reverse to fill in missing CARRIER_NAME values.

```
# Function 1: impute CARRIER from CARRIER_NAME
def impute_carrier(df):
    df = df.copy()
    name_to_carrier = (
        df.dropna(subset=['CARRIER', 'CARRIER_NAME'])
        .drop_duplicates('CARRIER_NAME')
        .set_index('CARRIER_NAME')['CARRIER']
    )
    carrier_to_name = (
        df.dropna(subset=['CARRIER', 'CARRIER_NAME'])
        .drop_duplicates('CARRIER')
        .set_index('CARRIER')['CARRIER_NAME']
    )

    mask = df['CARRIER'].isnull() & df['CARRIER_NAME'].notna()
    df.loc[mask, 'CARRIER'] = df.loc[mask, 'CARRIER_NAME'].map(name_to_carrier)

    mask2 = df['CARRIER_NAME'].isnull() & df['CARRIER'].notna()
    df.loc[mask2, 'CARRIER_NAME'] = df.loc[mask2, 'CARRIER'].map(carrier_to_name)

    print(f'CARRIER nulls remaining: {df["CARRIER"].isnull().sum()}')
    print(f'CARRIER_NAME nulls remaining: {df["CARRIER_NAME"].isnull().sum()}')
    return df
```

```
df = impute_carrier(df)
```

```
CARRIER nulls remaining: 0
CARRIER_NAME nulls remaining: 0
```

✓ 1.2 AIRLINE ID

```
df[['CARRIER', 'CARRIER_NAME', 'AIRLINE_ID']].drop_duplicates('CARRIER').head(10)
```

	CARRIER	CARRIER_NAME	AIRLINE_ID	
0	16	PSA Airlines Inc.	20397.0	
49	3Z	Tatonduk Flying Service	20408.0	
76	5X	United Parcel Service	19917.0	
351	5Y	Atlas Air Inc.	20007.0	
374	8C	Air Transport International	19874.0	
393	9E	Pinnacle Airlines Inc.	20363.0	
517	9S	Southern Air Inc.	20190.0	
525	AA	American Airlines Inc.	19805.0	
1255	ABX	ABX Air, Inc.	20453.0	
1319	AJQ	Aerodynamics Inc.	20354.0	

Each CARRIER code corresponds to exactly one AIRLINE_ID. Where CARRIER is known, AIRLINE_ID can be looked up from the same dataset.

```
# Function 2: impute AIRLINE_ID from CARRIER
def impute_airline_id(df):
    df = df.copy()
    lookup = (
        df.dropna(subset=['CARRIER', 'AIRLINE_ID'])
        .drop_duplicates('CARRIER')
        .set_index('CARRIER')['AIRLINE_ID']
    )
    mask = df['AIRLINE_ID'].isnull() & df['CARRIER'].notna()
    df.loc[mask, 'AIRLINE_ID'] = df.loc[mask, 'CARRIER'].map(lookup)
    print(f'AIRLINE_ID nulls remaining: {df["AIRLINE_ID"].isnull().sum()}')
    return df
```

```
df = impute_airline_id(df)
```

```
AIRLINE_ID nulls remaining: 0
```

1.3 MANUFACTURE_YEAR

```
print(df['MANUFACTURE_YEAR'].isnull().sum(), 'missing')
print(df['MANUFACTURE_YEAR'].describe())
```

```
3 missing
count    132310.000000
mean      2000.399992
std        39.630346
min         0.000000
25%       1995.000000
50%       2001.000000
75%       2007.000000
max       2756.000000
Name: MANUFACTURE_YEAR, dtype: float64
```

Only 3 values missing out of 132,313 rows. The year an aircraft was manufactured is not related to whether it is missing because there is no reason to think older or newer planes are systematically more likely to have a missing year. The missingness most plausibly depends on the aircraft MODEL, making this missing at random.

So for the imputation method, I used median MANUFACTURE_YEAR within the same MODEL group. Aircraft of the same model were built in similar periods, so the median gives a plausible value. I used Median over mean in case a model group has outlier years.

```
# Function 3: impute MANUFACTURE_YEAR with median (Missing at Random)
def impute_manufacture_year(df):
    df = df.copy()
    imputer = SimpleImputer(strategy='median')
    df['MANUFACTURE_YEAR'] = imputer.fit_transform(df[['MANUFACTURE_YEAR']])
    df['MANUFACTURE_YEAR'] = df['MANUFACTURE_YEAR'].round().astype('Int64')
    print(f'MANUFACTURE_YEAR nulls remaining: {df["MANUFACTURE_YEAR"].isnull().sum()}')
    return df
```

Double-click (or enter) to edit

```
df = impute_manufacture_year(df)
```

```
MANUFACTURE_YEAR nulls remaining: 0
```

1.4 NUMBER_OF_SEATS & CAPACITY_IN_POUNDS

```
print('NUMBER_OF_SEATS missing:', df['NUMBER_OF_SEATS'].isnull().sum())
print('CAPACITY_IN_POUNDS missing:', df['CAPACITY_IN_POUNDS'].isnull().sum())
print(df['NUMBER_OF_SEATS'].describe())
print(df['CAPACITY_IN_POUNDS'].describe())
```

```
NUMBER_OF_SEATS missing: 13238
CAPACITY_IN_POUNDS missing: 13332
count    119075.000000
mean      95.389494
std       63.414606
min        0.000000
25%       50.000000
50%      108.000000
75%      150.000000
max      190.000000
Name: NUMBER_OF_SEATS, dtype: float64
count    118981.000000
mean     69086.334600
std     84415.162714
min        0.000000
25%     34100.000000
50%     42380.000000
75%     78730.000000
max    875000.000000
Name: CAPACITY_IN_POUNDS, dtype: float64
```

Aircraft within the same model have very similar seat counts and payload capacities, as these are set by FAA type certification. There is no clear evidence that the magnitude of these values affects whether they are missing, so MNAR is unlikely; the missingness is more likely due to reporting gaps for certain models.

For imputation, I used the median within each MODEL group. Since both variables are right-skewed, the median provides a more robust estimate than the mean.

```
# Function 4: impute NUMBER_OF_SEATS and CAPACITY_IN_POUNDS with median by MODEL (MAR)
def impute_seats_and_capacity(df):
    df = df.copy()
    imputer = SimpleImputer(strategy='median')
    for col in ['NUMBER_OF_SEATS', 'CAPACITY_IN_POUNDS']:
        df[[col]] = imputer.fit_transform(df[[col]])
        print(f'{col} nulls remaining: {df[col].isnull().sum()}')
    return df
```

```
df = impute_seats_and_capacity(df)
```

```
NUMBER_OF_SEATS nulls remaining: 0
CAPACITY_IN_POUNDS nulls remaining: 0
```

```
df[cols_of_interest].isnull().sum()
```

```

      0
CARRIER      0
CARRIER_NAME  0
AIRLINE_ID     0
MANUFACTURE_YEAR  0
NUMBER_OF_SEATS  0
CAPACITY_IN_POUNDS  0
```

```
dtype: int64
```

TASK 2 - STADARDIZATION

2.1 Standardize MANUFACTURER

```
df['MANUFACTURER'].value_counts().head(25)
```

	count
MANUFACTURER	
BOEING	15922
Embraer	11508
THEBOEINGCO	9223
Bombardier	8871
Boeing	8392
BoeingCo	7446
AIRBUS	7179
AirbusIndustries	6967
BOEINGCOMPANY	6767
Airbus	5289
CESSNA	4157
EMBRAER	3287
MCDONNELL-DOUGLAS	3160
BOMBARDIER	2821
BOEING	2811
MCDONNELLDUGLAS	2781
CANADAIR	2734
AirbusIndustrie	2666
THEBOEINGCOMPANY	2142
TheBoeingCompany	1833
McDonnellDouglas	1525
GE	1110
ATR	1103
Douglas	990
AIRBUS	963

dtype: int64

The data contains many different versions of the same manufacturer name, for example, “boeing,” “THE BOEING COMPANY,” and “boeing co” all refer to the same company, so these need to be standardized. I first clean the text by lowercasing, trimming whitespace, and removing punctuation, then use an alias dictionary to map the most common variants to consistent labels.

```
# Function 5: standardize MANUFACTURER
def standardize_manufacturer(df):
    df = df.copy()
    print('Top values BEFORE:')
    print(df['MANUFACTURER'].value_counts().head(25))

    df['MANUFACTURER'] = df['MANUFACTURER'].astype(str).str.strip().str.lower().str.replace(r'^a-z\s', '', regex=True)

    alias = {
        # Boeing
        'boeing': 'Boeing',
        'boeing co': 'Boeing',
        'boeing company': 'Boeing',
        'the boeing company': 'Boeing',
        'boeingco': 'Boeing',
        'boeingcompany': 'Boeing',
        'theboeingco': 'Boeing',
        'theboeingcompany': 'Boeing',
        'boeingmcdonnelldouglas': 'Boeing',

        # Airbus
        'airbus': 'Airbus',
        'airbus industrie': 'Airbus',
        'airbus industries': 'Airbus',
        'airbus sas': 'Airbus',
        'airbusindustries': 'Airbus',
        'airbusindustrie': 'Airbus',
        'airbuscompany': 'Airbus',
    }
```

```

# Bombardier
'bombardier': 'Bombardier',
'bombardier inc': 'Bombardier',
'bombardier inc.': 'Bombardier',
'bombardieraerospace': 'Bombardier',

# Embraer
'embraer': 'Embraer',
'embraer sa': 'Embraer',
'embraer s a': 'Embraer',

# ATR
'atr': 'ATR',
'avions de transport regional': 'ATR',

# Cessna
'cessna': 'Cessna',
'cessna aircraft co': 'Cessna',
'cessna aircraft company': 'Cessna',

# McDonnell Douglas
'mcdonnell douglas': 'McDonnell Douglas',
'mcdonnell-douglas': 'McDonnell Douglas',
'mcdonnell Douglas': 'McDonnell Douglas',
'mcdonnell Douglas': 'McDonnell Douglas',
'douglas': 'McDonnell Douglas',
'douglas aircraft': 'McDonnell Douglas',
'mcdo': 'McDonnell Douglas',

# De Havilland
'dehavilland': 'De Havilland',
'de havilland': 'De Havilland',

# GE
'ge': 'GE',
'general electric': 'GE',

#Saab
'saab': 'Saab',
'saab aircraft': 'Saab',
}

df['MANUFACTURER'] = df['MANUFACTURER'].map(lambda x: alias.get(x, x.title()))

print('\nTop values AFTER:')
print(df['MANUFACTURER'].value_counts().head(25))
return df

```

```
df = standardize_manufacturer(df)
```

```

Top values BEFORE:
MANUFACTURER
BOEING                15922
Embraer              11508
THEBOEINGCO           9223
Bombardier            8871
Boeing                8392
BoeingCo              7446
AIRBUS                7179
AirbusIndustries      6967
BOEINGCOMPANY         6767
Airbus                5289
CESSNA                4157
EMBRAER               3287
MCDONNELL-DOUGLAS     3160
BOMBARDIER            2821
BOEING                2811
MCDONNELLDUGLAS       2781
CANADAIR              2734
AirbusIndustrie       2666
THEBOEINGCOMPANY      2142
TheBoeingCompany      1833
McDonnellDouglas      1525
GE                    1110
ATR                   1103
Douglas                990
AIRBUS                 963
Name: count, dtype: int64

Top values AFTER:
MANUFACTURER
Boeing                55639
Airbus                23513
Embraer               15554

```

```

Bombardier      12483
McDonnell Douglas 10742
Cessna          4514
Canadair        3861
ATR             1181
GE              1110
De Havilland    1084
Saab            595
Gulfstream      441
Convair         208
Lockheed        158
Dassault        127
Dornier         100
Curtiss         46
Dassault Sud    45
Easyjetgoal     44
Cebupacific     43
Piper           36
Challenger      36
Mcdonnell Douglas 30
Easyjetgecas    30
Iberiasmbc      28
Name: count, dtype: int64

```

2.2 Standardize MODEL

```
df['MODEL'].value_counts().head(25)
```

MODEL	count
EMB-145	2614
B-737-7H4	2470
B737-823	2370
A320-232	2333
A321-231	2259
737-700PASSENGERONLY	2027
C-208B	1872
B757-2	1775
CRJ-2/4	1761
B737-800PAX	1621
MD-80	1610
ERJ-170-200LR	1379
757-200	1345
CRJ200-2B19	1342
A319	1267
B-737-8H4	1256
CRJ-200	1148
ERJ-175	1132
SUPER80PASSENGER	1108
A320-1/2	1107
757-24APF	1039
B737-3	1036
MD-88-PSGR	1028
C-208B/3	1017
MD-11	991

```
dtype: int64
```

The output shows inconsistent casing and spacing (for example, "737-800" vs "737 800" vs "B737-800"). Standardization is needed so the same aircraft model is not counted as multiple different models.

The method is to strip whitespace, convert to upper-case, and collapse multiple spaces.

```
# Function 6: standardize MODEL
def standardize_model(df):
    df = df.copy()
    df['MODEL'] = (
        df['MODEL'].astype(str).str.strip().str.upper()
        .str.replace(r'\s+', ' ', regex=True)
    )
    print(f'Distinct models after cleaning: {df["MODEL"].nunique()}')
    return df
```

```
df = standardize_model(df)
```

```
Distinct models after cleaning: 1269
```

2.3 Standardize AIRCRAFT_STATUS

```
df['AIRCRAFT_STATUS'].value_counts(dropna=False)
```

	count
AIRCRAFT_STATUS	
O	79487
b	30852
B	12699
a	7804
A	1330
L	122
o	19

```
dtype: int64
```

The value_counts() output shows letter codes with mixed casing: "O", "o", "B", "b", "A", "a", "L". The same status appears in both upper and lower case, so case normalization is needed.

```
# Function 7: standardize AIRCRAFT_STATUS
def standardize_aircraft_status(df):
    df = df.copy()
    df['AIRCRAFT_STATUS'] = df['AIRCRAFT_STATUS'].astype(str).str.strip().str.lower()
    print(df['AIRCRAFT_STATUS'].value_counts())
    return df
```

```
df = standardize_aircraft_status(df)
```

```
AIRCRAFT_STATUS
o    79506
b    43551
a     9134
l      122
Name: count, dtype: int64
```

2.4 Standardize OPERATING_STATUS

```
df['OPERATING_STATUS'].value_counts(dropna=False)
```

	count
OPERATING_STATUS	
Y	126577
N	5664
y	71
	1

```
dtype: int64
```

Values are Y/N with one lowercase "y" and one blank " " entry. "Y" and "y" mean the same thing so we standardize to readable labels. The blank entry will be treated as unknown.


```
# Function 8: standardize OPERATING_STATUS
def standardize_operating_status(df):
    df = df.copy()
    df['OPERATING_STATUS'] = df['OPERATING_STATUS'].astype(str).str.strip().str.upper()
    op_map = {'Y': 'Operating', 'N': 'Not Operating'}
    df['OPERATING_STATUS'] = df['OPERATING_STATUS'].map(op_map).fillna('Unknown')
    print(df['OPERATING_STATUS'].value_counts())
    return df
```

```
df = standardize_operating_status(df)
```

```
OPERATING_STATUS
Operating      126648
Not Operating    5664
Unknown         1
Name: count, dtype: int64
```

Task 3 — Drop Remaining Missing

```
# Function 9: drop remaining missing rows
def drop_remaining_missing(df):
    before = len(df)
    df = df.dropna()
    after = len(df)
    print(f'Rows removed: {before - after:,}')
    print(f'Rows remaining: {after:,}')
    return df
```

```
df = drop_remaining_missing(df)
```

```
Rows removed: 31,026
Rows remaining: 101,287
```

Task 4 — Transformation and derivative variables

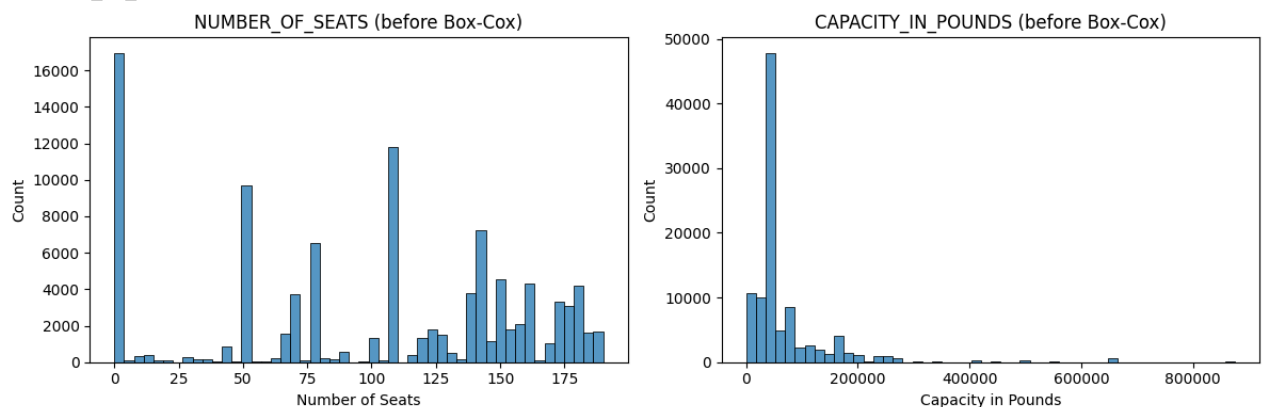
```
#check skewness and plot before transformation
print('NUMBER_OF_SEATS skewness: ', df['NUMBER_OF_SEATS'].skew().round(3))
print('CAPACITY_IN_POUNDS skewness: ', df['CAPACITY_IN_POUNDS'].skew().round(3))
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
```

```
sns.histplot(df['NUMBER_OF_SEATS'], bins=50, ax=axes[0])
axes[0].set_title('NUMBER_OF_SEATS (before Box-Cox)')
axes[0].set_xlabel('Number of Seats')
```

```
sns.histplot(df['CAPACITY_IN_POUNDS'], bins=50, ax=axes[1])
axes[1].set_title('CAPACITY_IN_POUNDS (before Box-Cox)')
axes[1].set_xlabel('Capacity in Pounds')
```

```
plt.tight_layout()
plt.show()
```

```
NUMBER_OF_SEATS skewness: -0.35
CAPACITY_IN_POUNDS skewness: 4.253
```



NUMBER_OF_SEATS (skewness = -0.35): The distribution has a large spike at 0 seats, which corresponds to cargo aircraft. Beyond that, it shows multiple peaks around 50, 75, and 140–150 seats, reflecting different aircraft types. Although the skewness is slightly negative, the main issue is the spike at 0 and the clearly non-normal shape.

CAPACITY_IN_POUNDS (skewness = 4.253): This distribution is strongly right-skewed. Most aircraft fall below 100,000 lbs, but a small number of very large aircraft create a long tail. The high skewness confirms this, making it a good candidate for a Box-Cox transformation.

```
# Function 10: apply Box-Cox transformation to NUMBER_OF_SEATS and CAPACITY_IN_POUNDS
from scipy.stats import boxcox
def apply_boxcox(df):
    df = df.copy()

    for col in ['NUMBER_OF_SEATS', 'CAPACITY_IN_POUNDS']:
        series = df[col].dropna()
        # Box-Cox requires strictly positive values
        if series.min() <= 0:
            series = series + abs(series.min()) + 1

        transformed, lam = boxcox(series)
        new_col = f'{col}_BOXCOX'
        df.loc[df[col].notna(), new_col] = transformed
        print(f'{col}: lambda = {lam:.4f}')

    return df
```

```
df = apply_boxcox(df)
```

```
NUMBER_OF_SEATS: lambda = 0.6872
CAPACITY_IN_POUNDS: lambda = 0.2964
```

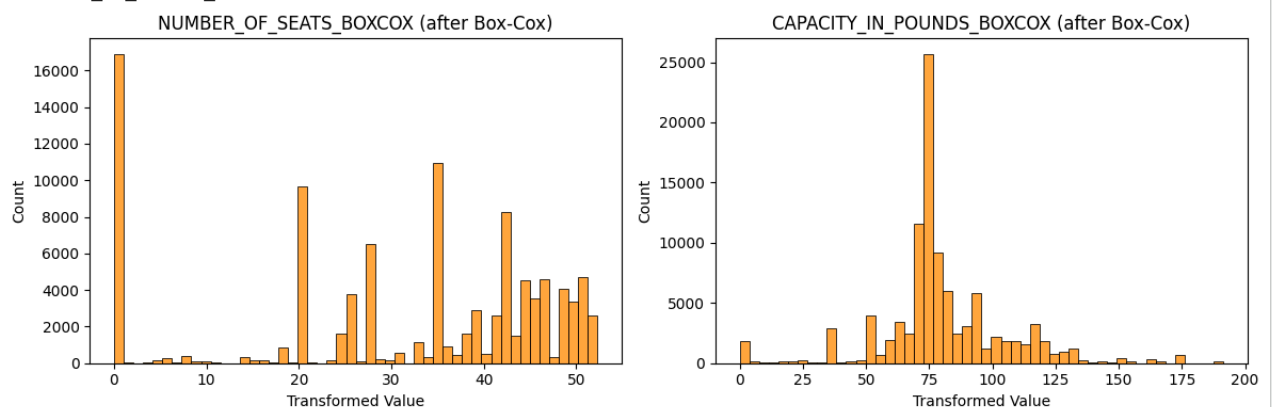
```
#check skewness and plot after transformation
print('NUMBER_OF_SEATS_BOXCOX skewness: ', df['NUMBER_OF_SEATS_BOXCOX'].skew().round(3))
print('CAPACITY_IN_POUNDS_BOXCOX skewness: ', df['CAPACITY_IN_POUNDS_BOXCOX'].skew().round(3))
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

sns.histplot(df['NUMBER_OF_SEATS_BOXCOX'], bins=50, ax=axes[0], color='darkorange')
axes[0].set_title('NUMBER_OF_SEATS_BOXCOX (after Box-Cox)')
axes[0].set_xlabel('Transformed Value')

sns.histplot(df['CAPACITY_IN_POUNDS_BOXCOX'], bins=50, ax=axes[1], color='darkorange')
axes[1].set_title('CAPACITY_IN_POUNDS_BOXCOX (after Box-Cox)')
axes[1].set_xlabel('Transformed Value')

plt.tight_layout()
plt.show()
```

```
NUMBER_OF_SEATS_BOXCOX skewness:    -0.714
CAPACITY_IN_POUNDS_BOXCOX skewness:  0.296
```



Observations after transformation:

NUMBER_OF_SEATS_BOXCOX (skewness = -0.714): The spike at 0 is still there, so the distribution remains uneven. This makes sense since cargo aircraft (0 seats) are a separate group, so Box-Cox doesn't really help here.

CAPACITY_IN_POUNDS_BOXCOX (skewness = 0.296): This improved a lot. The distribution is now much more symmetric, and the long right tail is mostly gone.

Summary: Box-Cox works well for CAPACITY_IN_POUNDS but not for NUMBER_OF_SEATS, since the seat data has a built-in split between cargo and passenger aircraft.

Task 5 — Feature Engineering & Modeling

```
# Function 11: create SIZE column
def create_size_column(df):
    df = df.copy()
    q1, q2, q3 = df['NUMBER_OF_SEATS'].quantile([0.25, 0.50, 0.75])
    print(f'Q1={q1}, Q2={q2}, Q3={q3}')

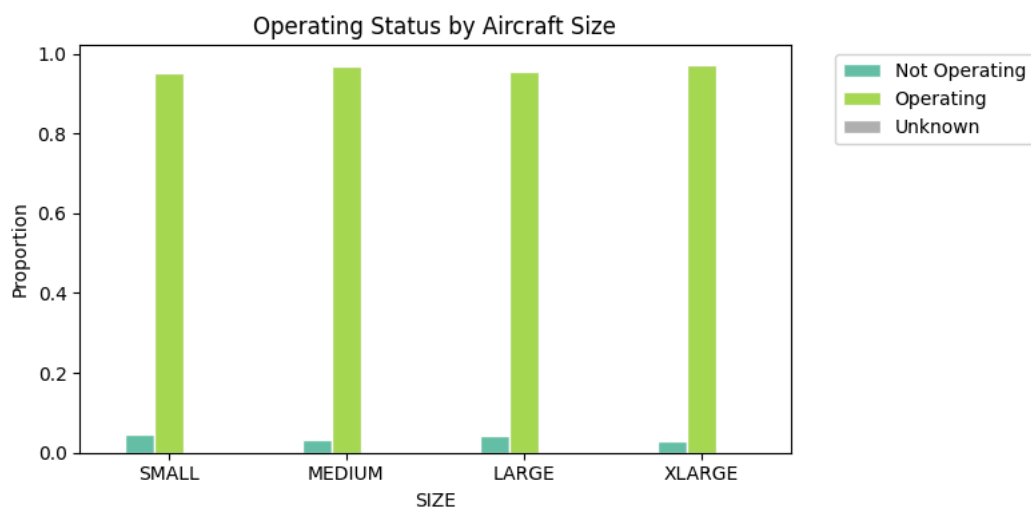
    bins = [-np.inf, q1, q2, q3, np.inf]
    labels = ['SMALL', 'MEDIUM', 'LARGE', 'XLARGE']
    df['SIZE'] = pd.cut(df['NUMBER_OF_SEATS'], bins=bins, labels=labels)
    print(df['SIZE'].value_counts())
    return df
```

```
df = create_size_column(df)
```

```
Q1=50.0, Q2=108.0, Q3=150.0
SIZE
SMALL    29323
MEDIUM   25313
LARGE    23461
XLARGE    23190
Name: count, dtype: int64
```

```
# operating status proportions by SIZE
size_order = ['SMALL', 'MEDIUM', 'LARGE', 'XLARGE']
op_prop = (
    df.groupby(['SIZE', 'OPERATING_STATUS'], observed=True)
      .size().unstack(fill_value=0)
      .reindex(size_order)
)
op_prop = op_prop.div(op_prop.sum(axis=1), axis=0)

op_prop.plot(kind='bar', figsize=(8, 4), colormap='Set2', edgecolor='white')
plt.title('Operating Status by Aircraft Size')
plt.ylabel('Proportion')
plt.xticks(rotation=0)
plt.legend(bbox_to_anchor=(1.05, 1))
plt.tight_layout()
plt.show()
```



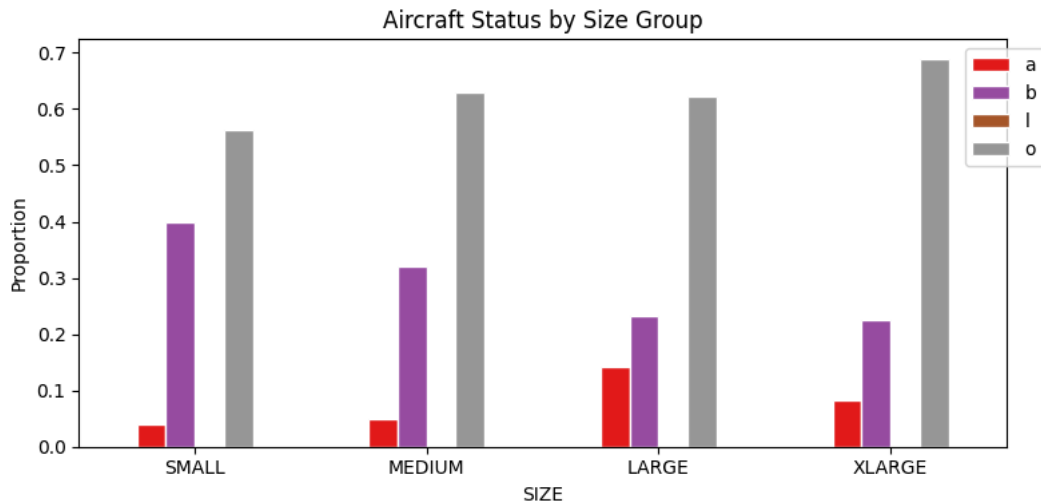
```
op_prop
```

OPERATING_STATUS	Not Operating	Operating	Unknown
SIZE			
SMALL	0.047062	0.952904	0.000034
MEDIUM	0.031920	0.968080	0.000000
LARGE	0.043647	0.956353	0.000000
XLARGE	0.027727	0.972273	0.000000

Next steps: [Generate code with op_prop](#) [New interactive sheet](#)

```
# aircraft status proportions by SIZE
ac_prop = (
    df.groupby(['SIZE', 'AIRCRAFT_STATUS'], observed=True)
      .size().unstack(fill_value=0)
      .reindex(size_order)
)
ac_prop = ac_prop.div(ac_prop.sum(axis=1), axis=0)

ac_prop.plot(kind='bar', figsize=(8, 4), colormap='Set1', edgecolor='white')
plt.title('Aircraft Status by Size Group')
plt.ylabel('Proportion')
plt.xticks(rotation=0)
plt.legend(bbox_to_anchor=(1.05, 1))
plt.tight_layout()
plt.show()
```



ac_prop

AIRCRAFT_STATUS	a	b	l	o	
SIZE					
SMALL	0.039321	0.398595	0.000000	0.562084	
MEDIUM	0.049619	0.320270	0.000514	0.629597	
LARGE	0.142108	0.232769	0.001961	0.623162	
XLARGE	0.081673	0.226261	0.002717	0.689349	

Next steps: [Generate code with ac_prop](#) [New interactive sheet](#)

Summary of Findings

Operating Status by Size: All size groups have high operating proportions (95–97%). XLARGE has the highest operating rate (97.2%) and SMALL the lowest (95.3%), with MEDIUM (96.8%) and LARGE (95.6%) in between.

Aircraft Status by Size: Out-of-service (o) is the dominant status across all size groups, increasing with size, SMALL (56.2%), MEDIUM (62.9%), LARGE (62.3%), and XLARGE (68.9%). Stored (b) is most common in smaller aircraft, SMALL (39.9%) and MEDIUM (32.0%) have notably higher stored proportions than LARGE (23.3%) and XLARGE (22.6%), suggesting smaller aircraft are more often temporarily parked rather than permanently retired. Active (a) proportion increases with size, LARGE (14.2%) and XLARGE (8.2%) are higher than SMALL (3.9%) and MEDIUM (5.0%), consistent with the operating status finding that larger aircraft are more likely to remain in active use.

Task 6 — Modeling refresher

```
# encode categoricals needed as features
df['SIZE_ENC'] = df['SIZE'].map({'SMALL': 0, 'MEDIUM': 1, 'LARGE': 2, 'XLARGE': 3})
df['MFR_ENC'] = LabelEncoder().fit_transform(df['MANUFACTURER'].fillna('Unknown'))
df['OP_ENC'] = (df['OPERATING_STATUS'] == 'Operating').astype(int)
```

```
feature_cols = ['MANUFACTURE_YEAR', 'AIRLINE_ID', 'SIZE_ENC', 'MFR_ENC', 'OP_ENC']

df_model = df[feature_cols + ['NUMBER_OF_SEATS', 'CAPACITY_IN_POUNDS']].dropna()
print(f'Rows available for modeling: {len(df_model):,}')

X = df_model[feature_cols]
```

Rows available for modeling: 101,287

6.1 Model 1 & 2 — Multivariate Regression:

```
# predict NUMBER_OF_SEATS with multivariate regression
y_seats = df_model['NUMBER_OF_SEATS']
X_train, X_test, y_train, y_test = train_test_split(X, y_seats, test_size=0.2, random_state=42)

lr_seats = LinearRegression()
lr_seats.fit(X_train, y_train)

rmse_train = root_mean_squared_error(y_train, lr_seats.predict(X_train))
rmse_test = root_mean_squared_error(y_test, lr_seats.predict(X_test))
print(f'Linear Regression - NUMBER_OF_SEATS')
print(f' Train RMSE: {rmse_train:.2f}')
print(f' Test RMSE: {rmse_test:.2f}')
```

Linear Regression - NUMBER_OF_SEATS
Train RMSE: 19.33
Test RMSE: 19.16

```
# predict CAPACITY_IN_POUNDS with multivariate regression
y_cap = df_model['CAPACITY_IN_POUNDS']
X_train, X_test, y_train, y_test = train_test_split(X, y_cap, test_size=0.2, random_state=42)

lr_cap = LinearRegression()
lr_cap.fit(X_train, y_train)

rmse_train = root_mean_squared_error(y_train, lr_cap.predict(X_train))
rmse_test = root_mean_squared_error(y_test, lr_cap.predict(X_test))
print(f'Linear Regression - CAPACITY_IN_POUNDS')
print(f' Train RMSE: {rmse_train:.2f}')
print(f' Test RMSE: {rmse_test:.2f}')
```

Linear Regression - CAPACITY_IN_POUNDS
Train RMSE: 87702.09
Test RMSE: 88684.37

6.2 Model 3 & 4 — Random Forest:

```
# predict NUMBER_OF_SEATS with random forest
y_seats = df_model['NUMBER_OF_SEATS']
X_train, X_test, y_train, y_test = train_test_split(X, y_seats, test_size=0.2, random_state=42)

rf_seats = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf_seats.fit(X_train, y_train)

rmse_train = root_mean_squared_error(y_train, rf_seats.predict(X_train))
rmse_test = root_mean_squared_error(y_test, rf_seats.predict(X_test))
print(f'Random Forest - NUMBER_OF_SEATS')
print(f' Train RMSE: {rmse_train:.2f}')
print(f' Test RMSE: {rmse_test:.2f}')
```

Random Forest - NUMBER_OF_SEATS
Train RMSE: 3.90
Test RMSE: 4.04

```
# predict CAPACITY_IN_POUNDS with random forest
y_cap = df_model['CAPACITY_IN_POUNDS']
X_train, X_test, y_train, y_test = train_test_split(X, y_cap, test_size=0.2, random_state=42)

rf_cap = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf_cap.fit(X_train, y_train)

rmse_train = root_mean_squared_error(y_train, rf_cap.predict(X_train))
rmse_test = root_mean_squared_error(y_test, rf_cap.predict(X_test))
print(f'Random Forest - CAPACITY_IN_POUNDS')
print(f' Train RMSE: {rmse_train:.2f}')
print(f' Test RMSE: {rmse_test:.2f}')
```

Random Forest - CAPACITY_IN_POUNDS
Train RMSE: 45770.61

Test RMSE: 47080.70

NUMBER_OF_SEATS: Random Forest performs much better than Linear Regression (test RMSE 4.04 vs 19.16), suggesting the relationship between features and seat count is non-linear. Train and test RMSE are very close for both models, so there is no sign of overfitting.

CAPACITY_IN_POUNDS: Random Forest again outperforms Linear Regression (test RMSE 47,080 vs 88,684). The larger absolute error values are expected given the scale of the variable (up to 875,000 lbs). Both models show a small and normal gap between train and test RMSE.

Overall: Random Forest is clearly the stronger model for both targets. All four models generalize well based on the small train-test